AQI PREDICTION PIPELINE

An End-to-End MLOps System for Real-Time Air Quality Monitoring and 72-Hour AQI Forecasting in Pakistan

INTERNSHIP PROJECT REPORT

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Abstract

Air pollution is an important environmental and public-health concern, and short-term forecasting can provide advanced information when air quality is expected to deteriorate. This internship project developed an end-to-end Machine Learning Operations (MLOps) system for real-time Air Quality Index (AQI) monitoring and 72-hour forecasting across ten Pakistani cities: Faisalabad, Hyderabad, Islamabad, Karachi, Lahore, Multan, Peshawar, Quetta, Rawalpindi, and Sialkot. The work covers data preparation, feature engineering, exploratory analysis, model development, explainability, production forecasting, monitoring, automation, API development, dashboard implementation, testing, and cloud deployment.

The processed historical dataset contains 43,170 observations and 33 columns covering January to June 2025. Weather and pollutant variables were combined with temporal, lag, and rolling AQI features, with Hopsworks incorporated for feature management. Seven conventional regression models were evaluated. LightGBM was selected as the production model with an RMSE of 6.7691, MAE of 4.1939, and R^2 of 0.9664. ARIMA and LSTM were also implemented as statistical and deep-learning experiments; their results are reported separately because their evaluation protocols differ from the deployed recursive forecasting pipeline.

The production system generates 24 predictions per city at three-hour intervals, producing a 72-hour forecast. It prefers accumulated real-time feature history when sufficient data is available and otherwise uses validated historical data as a fallback. SHAP provides global and local explanations. Monitoring evaluates feature drift and predictive performance, while conditional retraining logic avoids unnecessary model replacement. FastAPI exposes prediction services, GitHub Actions automates recurring workflows, and a Streamlit dashboard presents current AQI, weather and pollutant conditions, 72-hour forecasts, AQI categories,health guidance, hazardous-AQI alerts, local SHAP explanations, smart activity recommendations, and multi-city forecast comparison.The completed project demonstrates a production-oriented AQI forecasting system that integrates machine learning with practical MLOps engineering.

Keywords: Air Quality Index, AQI Forecasting, MLOps, LightGBM, LSTM, ARIMA, SHAP, Hopsworks, FastAPI, Streamlit, Pakistan

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1. Introduction

1.1 Background and Motivation

Air quality varies as pollutant concentrations interact with weather conditions and temporal patterns. Current AQI measurements describe present conditions but do not indicate how air quality may change during the following hours. Short-term forecasting can therefore support earlier awareness of unhealthy conditions.

This project treats AQI forecasting as an end-to-end engineering problem rather than an isolated modelling exercise. In addition to predictive accuracy, the system addresses data collection, feature consistency, model versioning, real-time inference, explainability, monitoring, automation, deployment, and user-facing communication.

1.2 Problem Statement

The project addresses the need for a reliable and maintainable system that can monitor environmental conditions and generate 72-hour AQI forecasts for major Pakistani cities. The system must preserve temporal context, use current environmental information, provide interpretable predictions, detect operational changes, and remain accessible through a deployed application.

1.3 Project Objectives

- Prepare and validate historical AQI, pollutant, and weather data.
- Build a real-time data and feature pipeline for ten Pakistani cities.
- Engineer temporal, lag, and rolling-window features.
- Compare conventional machine-learning models and evaluate statistical and deep-learning approaches.
- Select and version a production model and generate 72-hour forecasts at three-hour intervals.
- Implement SHAP explainability, drift monitoring, performance monitoring, and conditional retraining.
- Expose prediction functionality through FastAPI and a Streamlit dashboard.
- Automate recurring workflows with GitHub Actions and deploy the application.
- Provide AQI categories, health guidance, and hazardous-condition alerts.

2. Project Scope and Requirement Coverage

The internship required a detailed record of the work achieved. The implementation therefore covers the complete machine-learning lifecycle, from feature preparation and experimentation to production forecasting, monitoring, automation, testing, and deployment.

Area	Implemented Scope
Data pipeline	Historical and real-time weather/pollutant processing; temporal and derived AQI features; historical backfill and feature management.
Model development	Seven regression models plus ARIMA and LSTM experiments; evaluation using RMSE, MAE, and R^2.
Explainability	Global SHAP analysis and local prediction-specific SHAP explanations integrated into the dashboard.
Forecasting	72-hour recursive forecasts every three hours for ten cities.
MLOps	Model registry, monitoring, retraining decisions, GitHub Actions automation, and testing.

Area	Implemented Scope
Application	FastAPI services and Streamlit dashboard with current conditions, 72-hour forecasts, health guidance, alerts, local SHAP explanations, Smart Activity Planner, City Comparison, and CSV export.
Deployment	Public Streamlit Community Cloud deployment with secrets managed outside source code.

Table 1. Summary of implemented project scope.

3. System Architecture

The architecture separates historical model development from real-time production forecasting while keeping feature definitions and model artifacts consistent. Historical data supports EDA, training, evaluation, explainability, and fallback forecasting. Real-time observations are accumulated and transformed into model-ready features. The registered LightGBM model generates recursive forecasts, while monitoring and automation components support production operation.

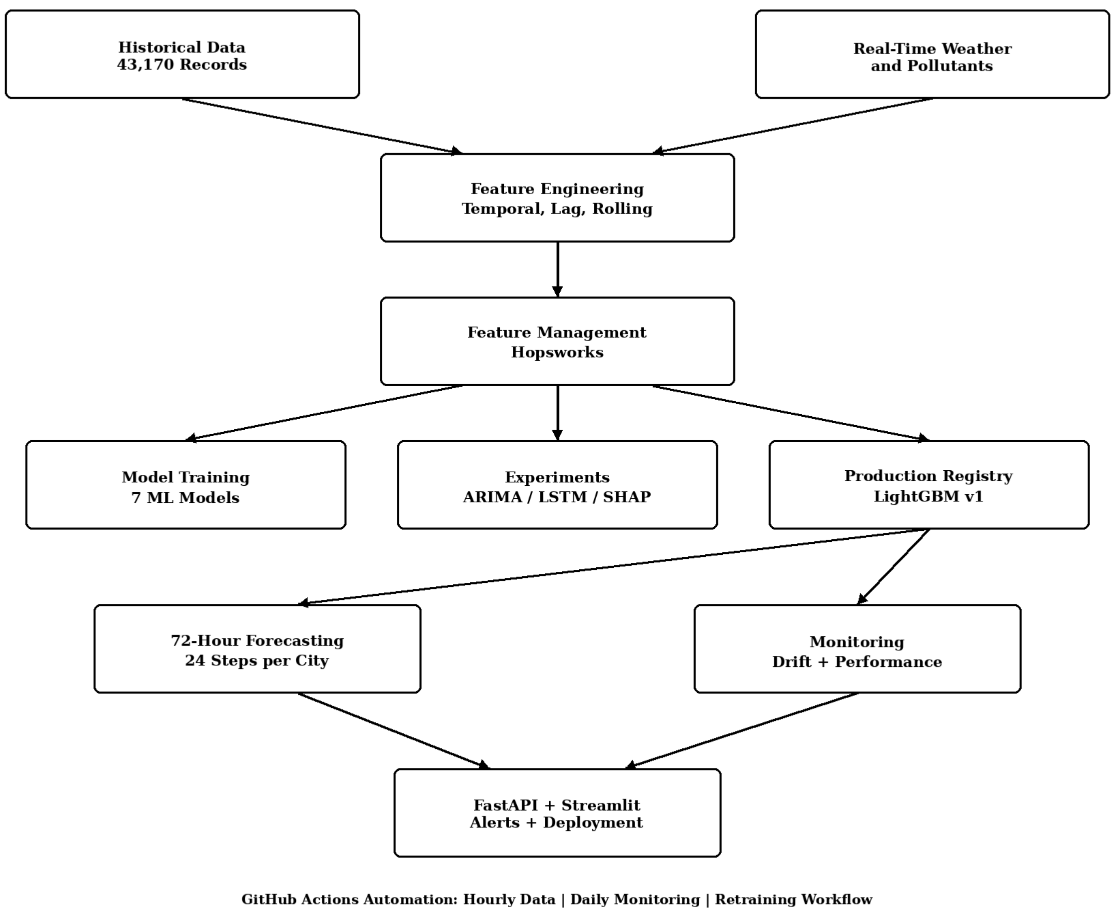


Figure 1. End-to-end architecture of the AQI prediction and MLOps pipeline.

3.1 Operational Data Flow

For each city, the forecasting pipeline checks whether sufficient real-time feature history is available. Valid real-time data is preferred; otherwise, validated historical features are used as a fallback. The model predicts one future AQI value at a time, and each predicted value contributes to subsequent lag and rolling calculations until 24 three-hour steps have been generated, covering 72 hours.

3.2 Supporting MLOps Components

Hopsworks provides feature management, a local versioned registry identifies the active production model, SHAP provides explainability, GitHub Actions automates recurring workflows, and monitoring combines feature drift with predictive performance to determine whether retraining is necessary.

4. Data Collection and Dataset

4.1 Historical Dataset

The processed model-training dataset contains 43,170 rows and 33 columns for ten Pakistani cities and covers 2 January 2025 to 30 June 2025. The supervised target is target_aqi. Validation found no missing values and no duplicate rows in the processed dataset.

Property	Value
Rows	43,170
Columns	33
Cities	10
Date range	02 January 2025 - 30 June 2025
Target	target_aqi
Missing values	None
Duplicate rows	0

Table 2. Historical dataset summary.

4.2 Feature Groups

The data combines location, weather, pollutant, temporal, and AQI-history variables. Environmental inputs include temperature, humidity, pressure, cloud cover, wind speed, wind direction, precipitation, PM10, PM2.5, carbon monoxide, nitrogen dioxide, sulphur dioxide, ozone, and current US AQI. Temporal variables include hour, day, month, year, day of week, weekend indicator, and season. Engineered history features include AQI lags at 1, 3, 6, and 24 periods and rolling AQI statistics over 3, 6, and 24 periods.

4.3 Real-Time Data Collection

Real-time weather and pollutant observations are collected and transformed into the structure required by the production model. The dataset accumulates history because lag and rolling features cannot be constructed from a single observation. A validated real-time run accumulated 350 records, with 35 observations per city and approximately 102 hours of coverage, which was sufficient for all ten cities to use the real-time forecasting path.

timestamp	city	temperature	humidity	pm2_5	pm10	ozone	us_aqi
2026-08-31 06:00:00	Faisalabad	39.68	22.0	40.21	130.13	91.23	113.0
2026-08-31 06:00:00	Hyderabad	27.23	65.0	1.83	2.72	32.52	8.0
2026-08-31 06:00:00	Islamabad	32.69	70.0	86.54	162.76	128.08	167.0
2026-08-31 06:00:00	Karachi	27.90	74.0	13.56	55.24	37.31	54.0
2026-08-31 06:00:00	Lahore	33.99	52.0	35.30	118.06	89.37	100.0
2026-08-31 06:00:00	Multan	31.94	62.0	34.98	128.67	72.66	99.0
2026-08-31 06:00:00	Peshawar	33.08	58.0	75.00	184.04	113.43	161.0
2026-08-31 06:00:00	Quetta	31.82	8.0	25.60	125.93	113.35	86.0
2026-08-31 06:00:00	Rawalpindi	32.91	70.0	86.54	162.76	128.08	167.0
2026-08-31 06:00:00	Sialkot	32.90	66.0	76.08	161.63	118.07	162.0

Figure 2. Evidence of real-time environmental data collection and feature accumulation.

4.4 Data Limitation Identified

Islamabad and Rawalpindi contain identical AQI values across 4,317 comparable timestamps, with a maximum difference of 0.00. The records were retained because they belong to the dataset used by the project, but the duplication is documented because it reduces the independence of city-level analysis.

5. Feature Engineering and Hopsworks

Feature engineering represents both current environmental conditions and recent AQI behavior. Temporal variables capture recurring daily and seasonal effects, while lag and rolling features provide short-term historical context. Pollutant and weather variables allow tree-based models to learn nonlinear relationships beyond AQI history alone.

5.1 Hopsworks Feature Management

Hopsworks is incorporated as the feature-management layer. The historical feature group aqi_features version 1 stores the engineered representation used by the machine-learning workflow. This creates a structured interface between data preparation and model development and supports consistent feature reuse.

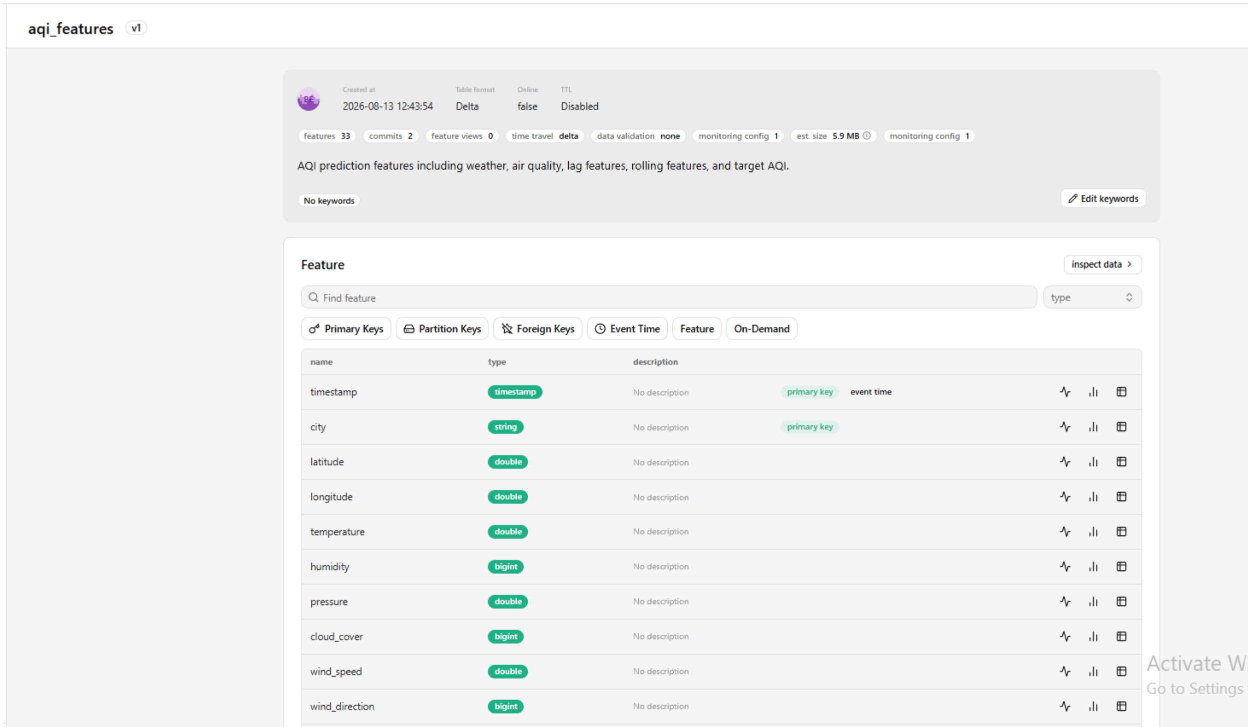
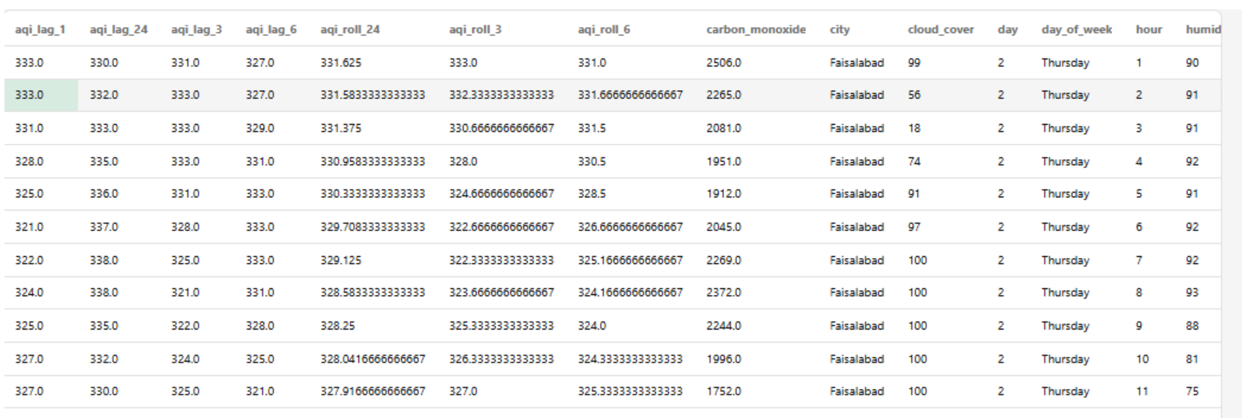


Figure 3. Hopsworks aqi_features v1 feature group containing environmental, temporal, lag, rolling, and target features used by the AQI prediction pipeline.



aqi_lag_1	aqi_lag_24	aqi_lag_3	aqi_lag_6	aqi_roll_24	aqi_roll_3	aqi_roll_6	carbon_monoxide	city	cloud_cover	day	day_of_week	hour	humid
333.0	330.0	331.0	327.0	331.625	333.0	331.0	2506.0	Faisalabad	99	2	Thursday	1	90
333.0	332.0	333.0	327.0	331.5833333333333	332.3333333333333	331.6666666666667	2265.0	Faisalabad	56	2	Thursday	2	91
331.0	333.0	333.0	329.0	331.375	330.6666666666667	331.5	2081.0	Faisalabad	18	2	Thursday	3	91
328.0	335.0	333.0	331.0	330.9583333333333	328.0	330.5	1951.0	Faisalabad	74	2	Thursday	4	92
325.0	336.0	331.0	333.0	330.3333333333333	324.6666666666667	328.5	1912.0	Faisalabad	91	2	Thursday	5	91
321.0	337.0	328.0	333.0	329.7083333333333	322.6666666666667	326.6666666666667	2045.0	Faisalabad	97	2	Thursday	6	92
322.0	338.0	325.0	333.0	329.125	322.3333333333333	325.1666666666667	2269.0	Faisalabad	100	2	Thursday	7	92
324.0	338.0	321.0	331.0	328.5833333333333	323.6666666666667	324.1666666666667	2372.0	Faisalabad	100	2	Thursday	8	93
325.0	335.0	322.0	328.0	328.25	325.3333333333333	324.0	2244.0	Faisalabad	100	2	Thursday	9	88
327.0	332.0	324.0	325.0	328.0416666666667	326.3333333333333	324.3333333333333	1996.0	Faisalabad	100	2	Thursday	10	81
327.0	330.0	325.0	321.0	327.9166666666667	327.0	325.3333333333333	1752.0	Faisalabad	100	2	Thursday	11	75
327.0	330.0	327.0	322.0	327.75	326.6666666666667	326.0	1465.0	Faisalabad	100	2	Thursday	12	67

6. Exploratory Data Analysis

EDA verified data quality and identified patterns relevant to forecasting. The analysis covered AQI distributions, city differences, monthly and hourly trends, feature correlations, AQI categories, and hazardous observations.

6.1 AQI Distribution and City Differences

The target AQI has a mean of 119.07, median of 110, standard deviation of 47.74, minimum of 33, and maximum of 538. The distribution is right-skewed, with most observations concentrated approximately between 60 and 180 and a smaller high-AQI tail.

City	Mean	Median	Min	Max
Faisalabad	160.24	153	70	437
Hyderabad	87.60	81	39	162
Islamabad	108.33	99	51	206
Karachi	87.59	81	52	163
Lahore	152.02	149	56	538
Multan	150.89	151	62	340
Peshawar	108.24	100	60	211
Quetta	87.57	77	33	319
Rawalpindi	108.33	99	51	206
Sialkot	140.47	139	56	303

Table 3. City-level historical AQI statistics.

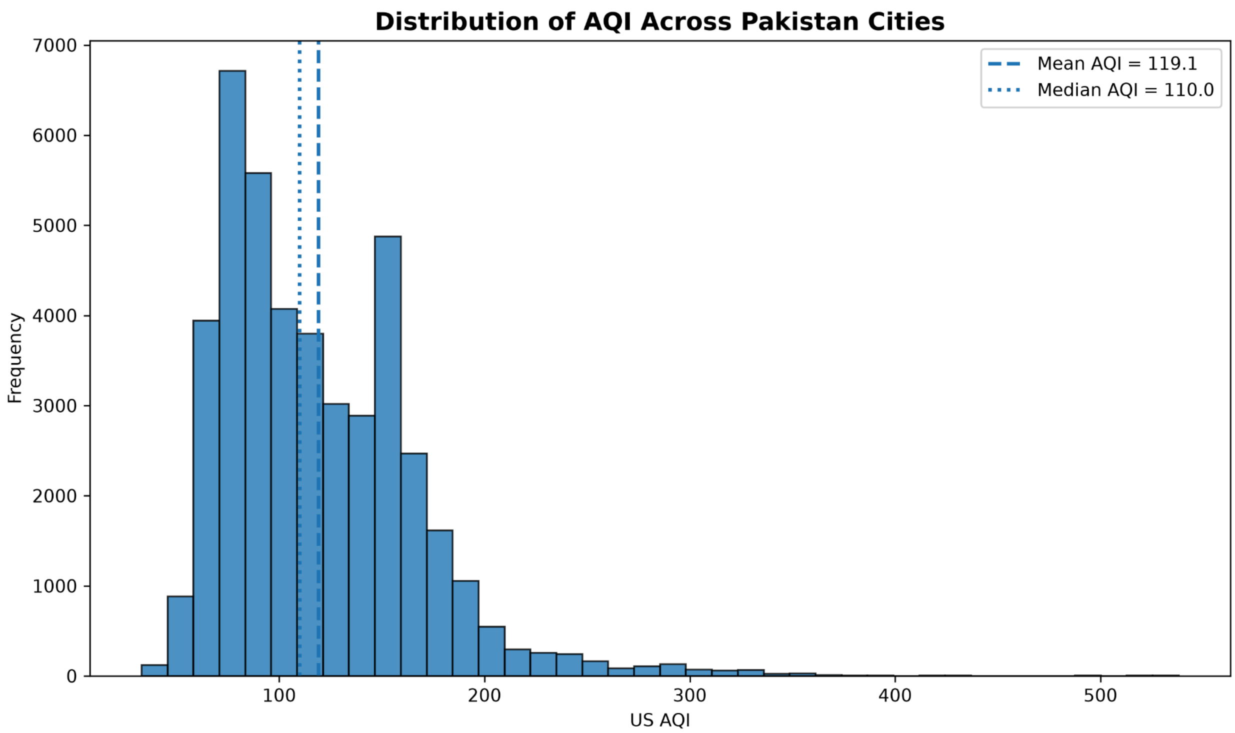


Figure 4. Distribution of target AQI.

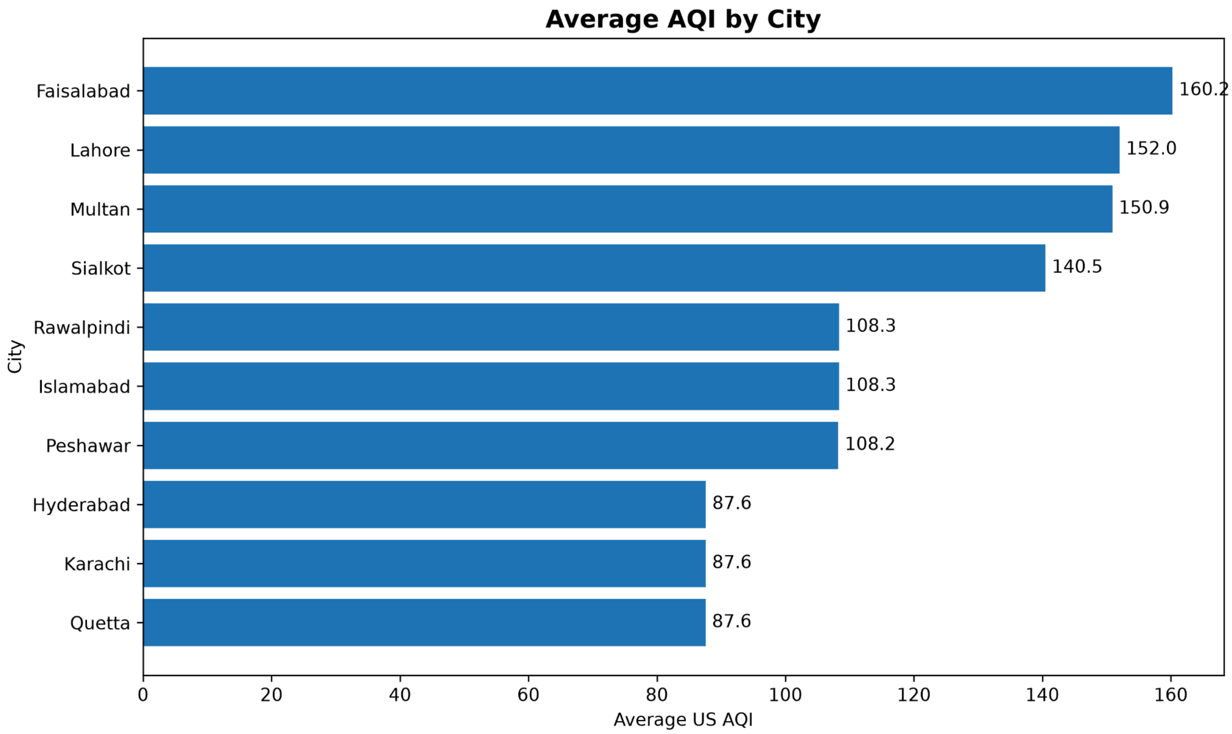


Figure 5. Average AQI by city.

6.2 Temporal Patterns

Monthly analysis shows higher January AQI in Faisalabad, Lahore, Multan, and Sialkot, followed by lower values during February and March and increases in several cities during May and June. Karachi and Hyderabad are comparatively lower and more stable, while Quetta follows a different pattern with increases during April and May.

Hourly AQI is relatively stable from midnight through late morning and rises during the afternoon. The highest average values occur around 18:00 and 19:00, at approximately 135.45 and 135.12 respectively.

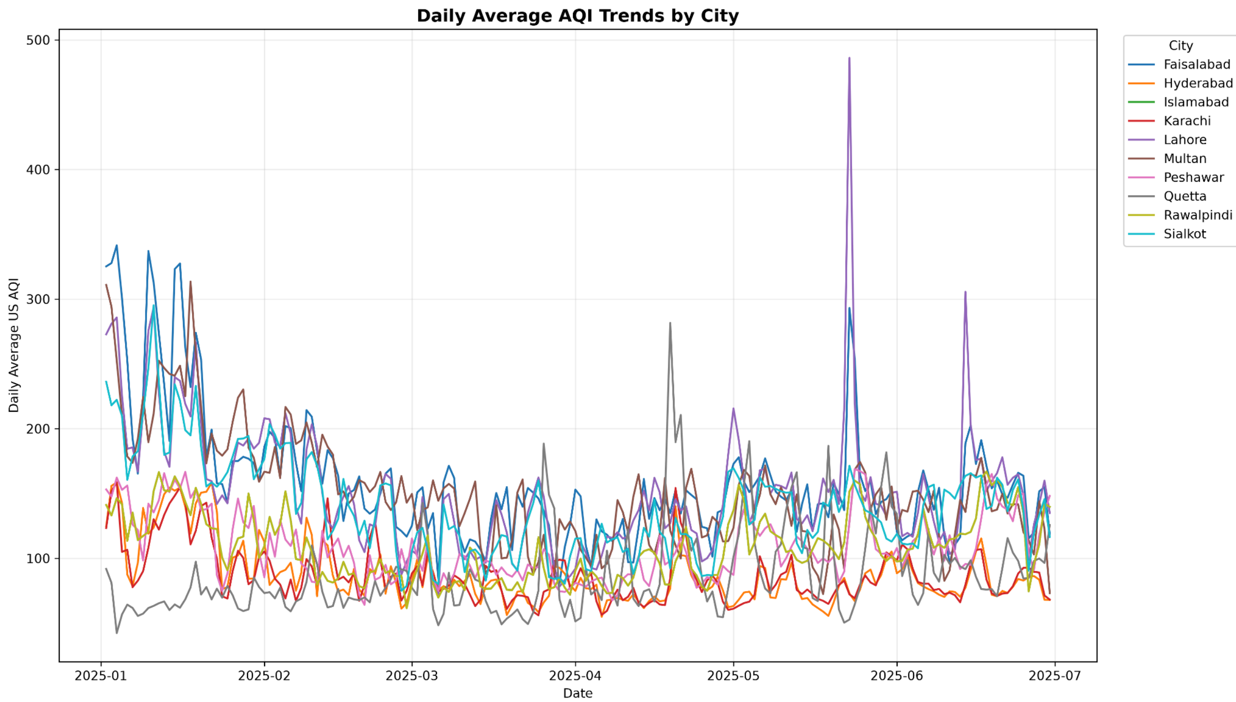


Figure 6. Historical AQI trends across cities.

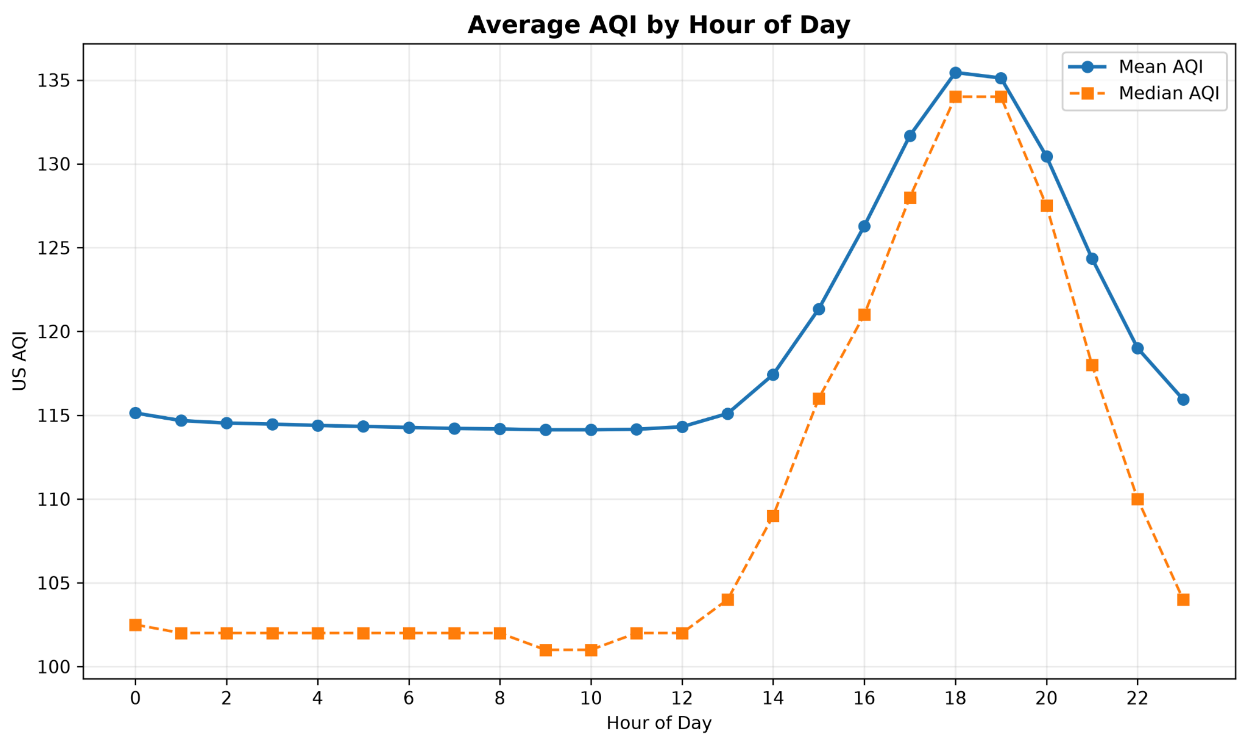


Figure 7. Average AQI by hour of day.

6.3 Feature Relationships

The strongest linear relationship with future AQI is current us_aqi at 0.9481, followed by rolling and lagged AQI features. PM2.5 is the pollutant with the strongest linear correlation to the target at 0.7194. Correlation describes association and is not interpreted as causality.

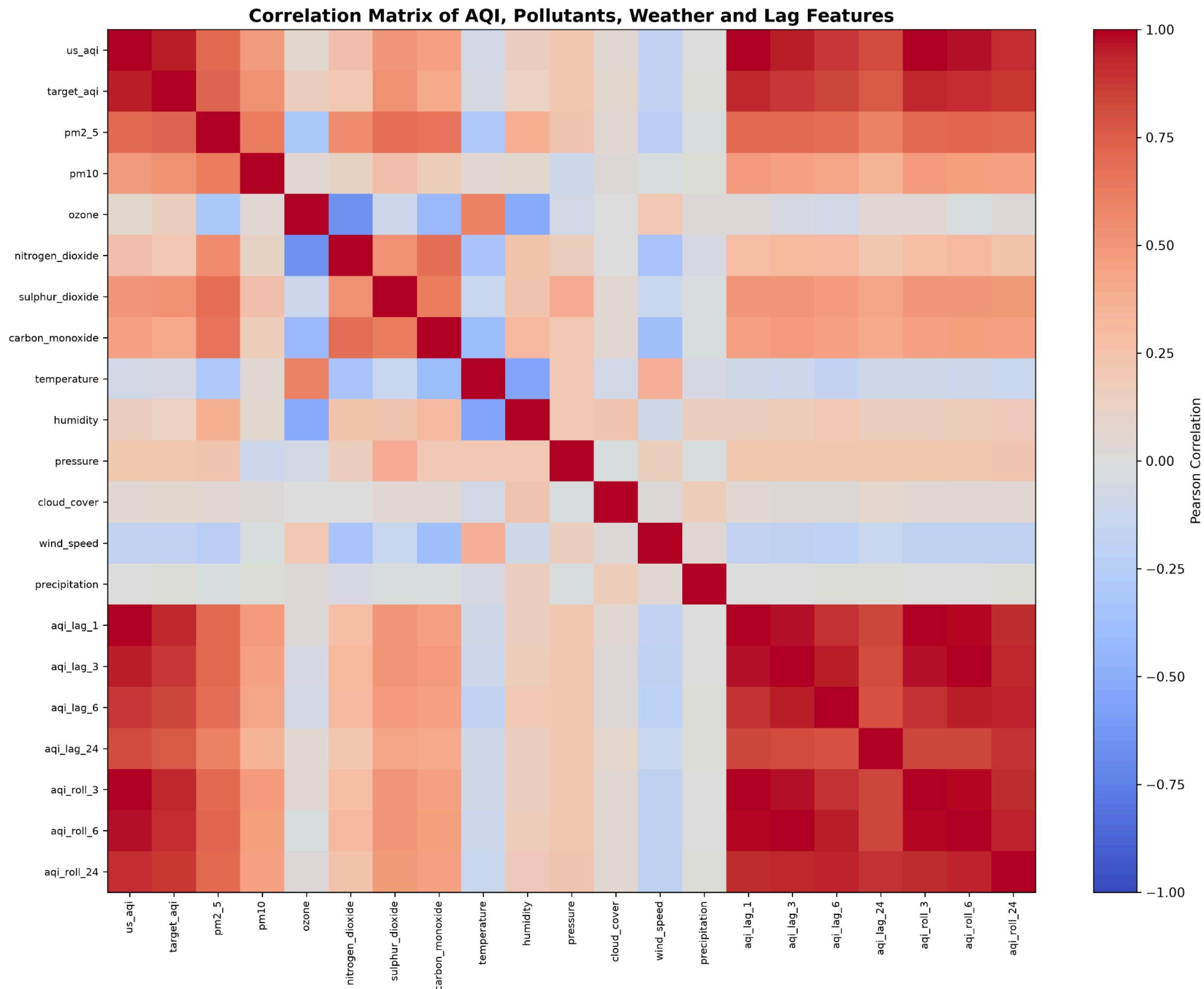


Figure 8. Correlation heatmap for numerical model features.

6.4 AQI Categories

Category	Observations	Percentage
Good	219	0.51%
Moderate	18,672	43.25%
Unhealthy for Sensitive Groups	13,127	30.41%
Unhealthy	9,197	21.30%
Very Unhealthy	1,664	3.85%
Hazardous	291	0.67%

Table 4. Historical AQI category distribution.

Overall, 55.82% of observations are above the Moderate category. Hazardous observations are concentrated mainly in Faisalabad (191), Lahore (45), and Multan (44), with smaller counts in Sialkot (6) and Quetta (5).

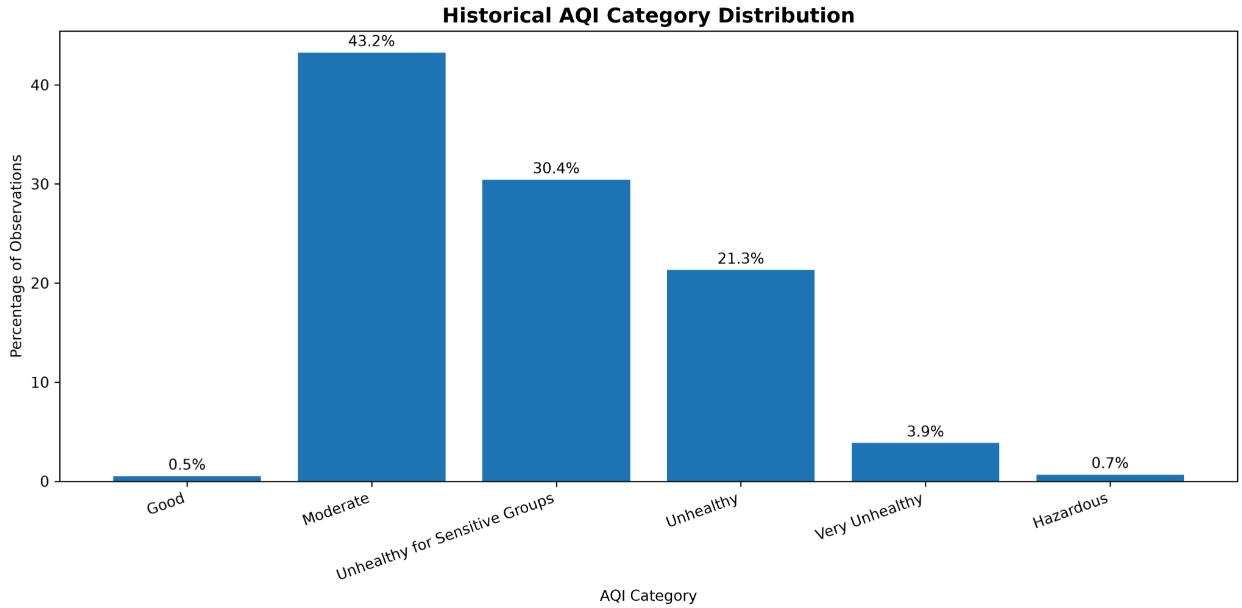


Figure 9. Distribution of historical AQI categories.

7. Machine-Learning Model Development and Evaluation

Seven conventional regression models were trained and evaluated. Model selection considered MAE, RMSE, and R^2, with RMSE used as the principal production ranking metric.

Rank	Model	MAE	RMSE	R^2
1	LightGBM	4.1939	6.7691	0.9664
2	ExtraTrees	4.0128	6.8341	0.9657
3	RandomForest	4.0226	6.9461	0.9646
4	GradientBoosting	4.4233	7.0672	0.9633
5	CatBoost	5.1222	7.7025	0.9564
6	XGBoost	4.3077	7.7887	0.9555
7	Ridge	7.9320	11.5007	0.9029

Table 5. Conventional machine-learning model comparison.

LightGBM achieved the lowest RMSE and was selected as the production model. Extra Trees produced a slightly lower MAE, but LightGBM provided the strongest RMSE while maintaining high overall fit and was integrated into the production workflow.

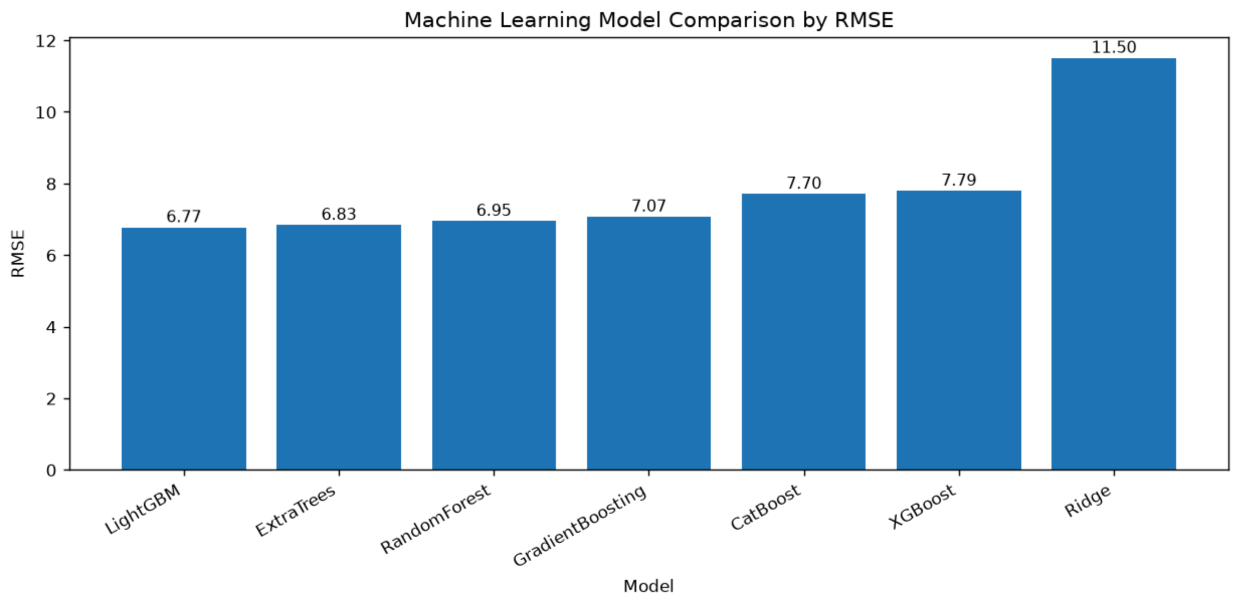


Figure 10. Comparison of conventional model RMSE values.

8. Statistical and Deep-Learning Forecasting Experiments

8.1 ARIMA Baseline

A univariate ARIMA(1,1,1) model was evaluated separately for each city using a chronological 80/20 split of target AQI. The combined result was MAE 30.6199, RMSE 39.8168, and R^2 -0.1641. The negative R^2 indicates that this simple long-horizon univariate baseline did not model the holdout period effectively. It also does not use the pollutant, weather, and engineered exogenous features available to the production model.

8.2 LSTM Experiment

A separate univariate LSTM sequence model was trained per city using a 24-step input sequence and chronological 80/20 split. The architecture contains an LSTM layer with 32 units followed by dense layers. Training used Adam optimization, mean squared error loss, early stopping, and no sequence shuffling.

The combined LSTM experiment achieved MAE 3.5063, RMSE 5.5521, and R^2 0.9774 under its one-step-ahead sequence evaluation. This result is not treated as a direct ranking against production LightGBM because the protocols differ. The LSTM uses observed AQI context for one-step-ahead testing, whereas the deployed LightGBM system performs recursive 72-hour forecasting using a richer feature set. LightGBM therefore remains the production model.

Approach	MAE	RMSE	R^2	Role
ARIMA(1,1,1)	30.6199	39.8168	-0.1641	Statistical baseline
LSTM	3.5063	5.5521	0.9774	Deep-learning experiment
LightGBM	4.1939	6.7691	0.9664	Production model; different evaluation workflow

Table 6. Summary of forecasting approaches. Metrics originate from their respective evaluation protocols.

9. Production Model Registry and 72-Hour Forecasting

9.1 Model Registry

The selected LightGBM model is stored in a local versioned registry. Registry metadata identifies the active production model, while the serialized artifact is stored under models/registry/LightGBM/v1/model.pkl. The active version is LightGBM v1 with registered RMSE 6.7691. This design separates model selection from application code and provides a foundation for promotion or rollback.


```json
{
    "models": {
        "LightGBM": {
            "production_version": "v1",
            "versions": [
                {
                    "model_name": "LightGBM",
                    "version": "v1",
                    "registered_at": "2026-08-22T10:01:43.525783+00:00",
                    "dataset_version": "2025_Jan_Jun",
                    "metrics": {
                        "MAE": 4.1939,
                        "RMSE": 6.7691,
                        "R2": 0.9664,
                        "MAPE": 0.0
                    },
```


Figure 11. Versioned production model registry.

9.2 Recursive Forecast Generation

The production pipeline generates 24 future points per city at three-hour intervals. After each prediction, AQI lag and rolling features are updated recursively so the next step can be constructed. Exogenous pollutant and weather values are persisted through the recursive horizon. A complete ten-city run produces 240 forecast rows.

After sufficient real-time history accumulated, all ten cities successfully switched to the REALTIME source. Forecast ranges remained city-specific, for example approximately 38.51-42.46 in Hyderabad, 65.87-72.55 in Karachi, 137.83-142.33 in Lahore, and 170.79-179.22 in Islamabad during the validated run.

```
            Forecast_Rows Data_Source
city
Faisalabad             24    REALTIME
Hyderabad              24    REALTIME
Islamabad              24    REALTIME
Karachi                24    REALTIME
Lahore                 24    REALTIME
Multan                 24    REALTIME
Peshawar               24    REALTIME
```

Figure 12. Successful real-time 72-hour forecast generation.

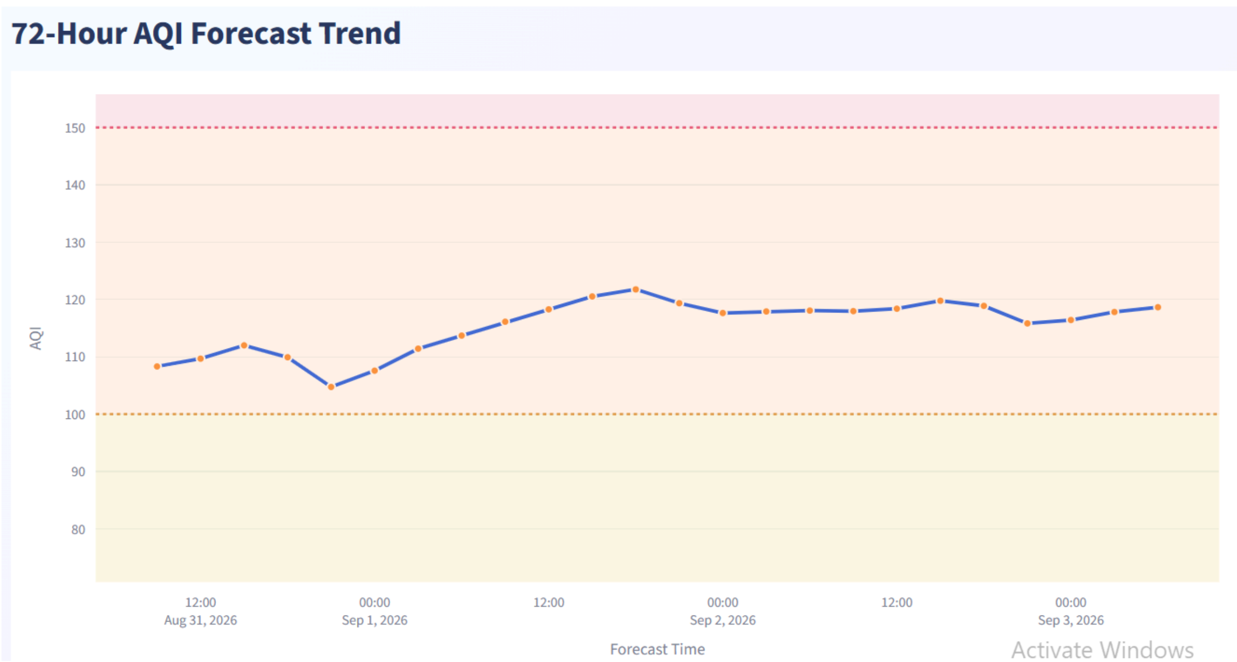


Figure 13. Example city-level 72-hour AQI forecast.

10. Model Explainability with SHAP

SHAP was implemented to explain the production LightGBM model. The explainer uses chronological evaluation data, applies the fitted preprocessing pipeline, and computes TreeExplainer SHAP values on a 2,000-row sample. The transformed input contains 47 model features.

Rank	Feature	Mean |SHAP|
1	us_aqi	27.0146
2	ozone	5.5398
3	hour	2.9071
4	pm2_5	2.4659
5	aqi_lag_1	1.1253
6	pm10	0.8134
7	aqi_roll_24	0.8116

Rank	Feature	Mean |SHAP|
8	aqi_lag_6	0.8060
9	nitrogen_dioxide	0.6035
10	temperature	0.6020

Table 7. Ten most influential features by mean absolute SHAP value.

Current AQI is the dominant feature, while ozone, hour, PM2.5, lagged AQI, PM10, and rolling AQI also contribute substantially. Ozone has relatively weak Pearson correlation with the target but high SHAP importance, which is consistent with a nonlinear tree model capturing relationships that simple linear correlation does not describe. SHAP values explain model contributions and do not establish causality.

The final dashboard extends SHAP from offline model analysis to user-facing prediction explanation. For the selected city, the dashboard reconstructs the model input corresponding to the next forecast step and calculates local SHAP contributions using the production LightGBM pipeline. Positive SHAP values indicate features that increase the predicted AQI, while negative values indicate features that reduce it. The dashboard displays the explained prediction, model baseline, strongest increasing factors, strongest reducing factor, and a detailed feature-contribution table. This allows users to understand why the next AQI prediction is high or low.

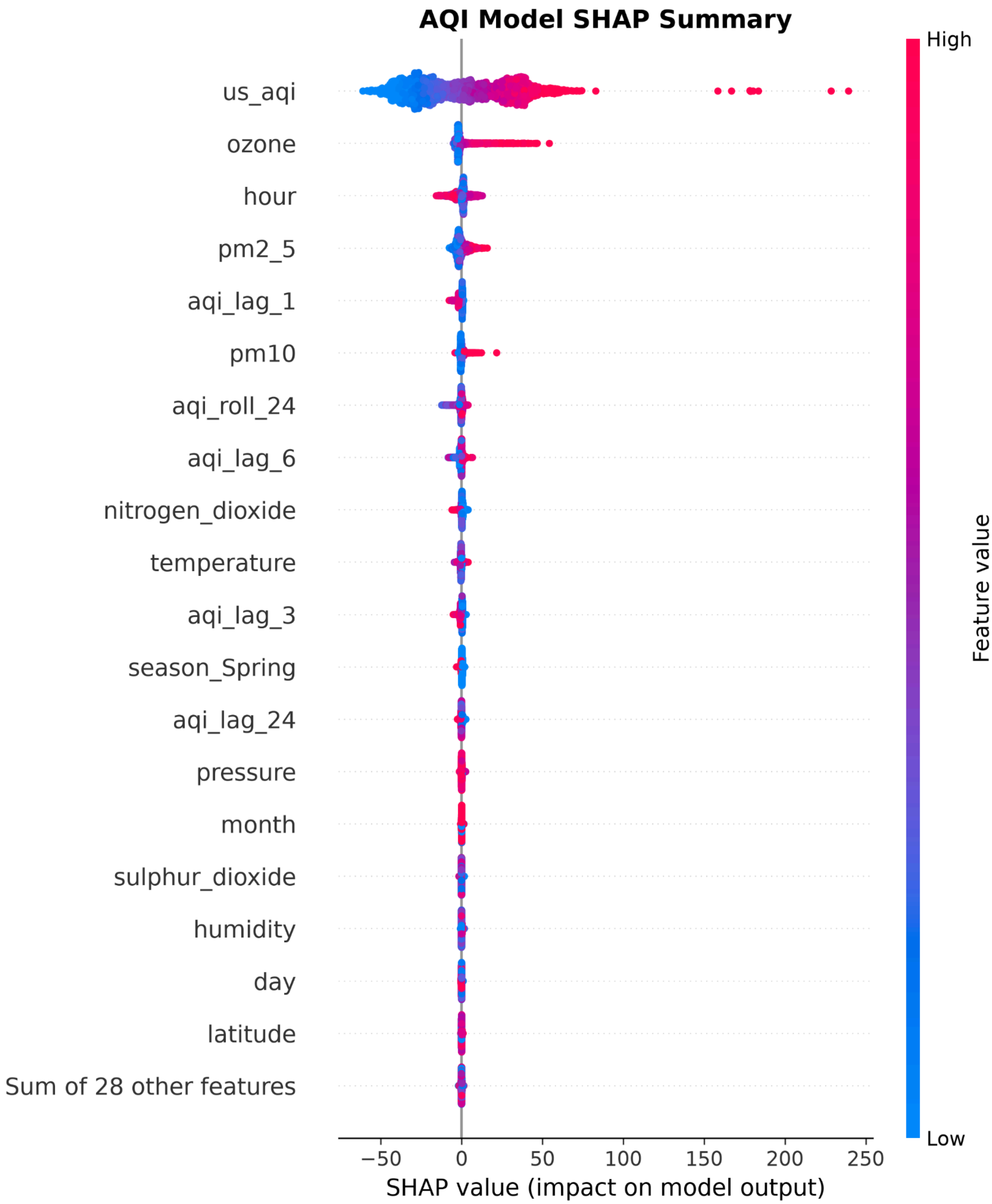


Figure 14. SHAP beeswarm summary for the LightGBM model.

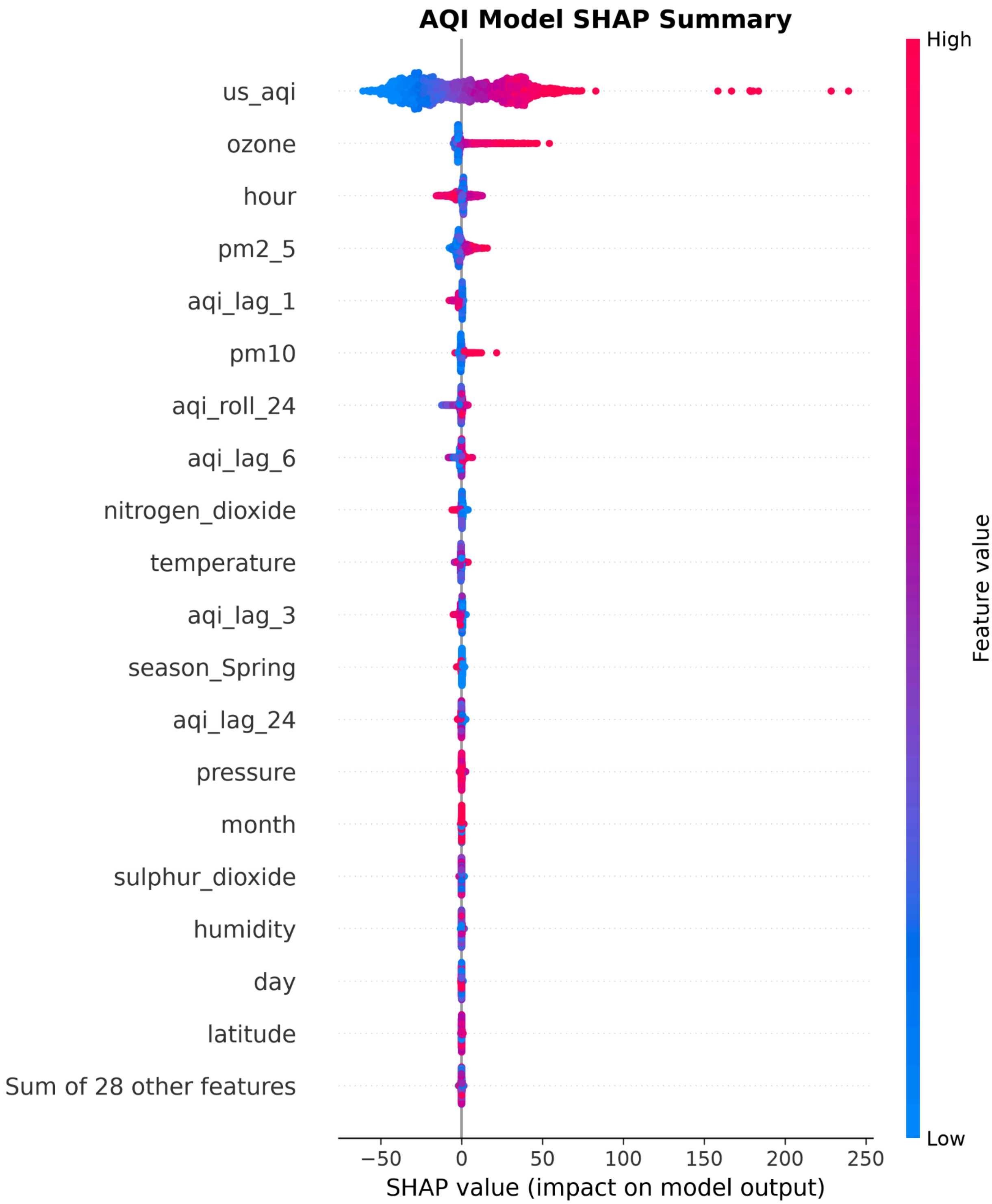


Figure 15. Global SHAP feature importance.

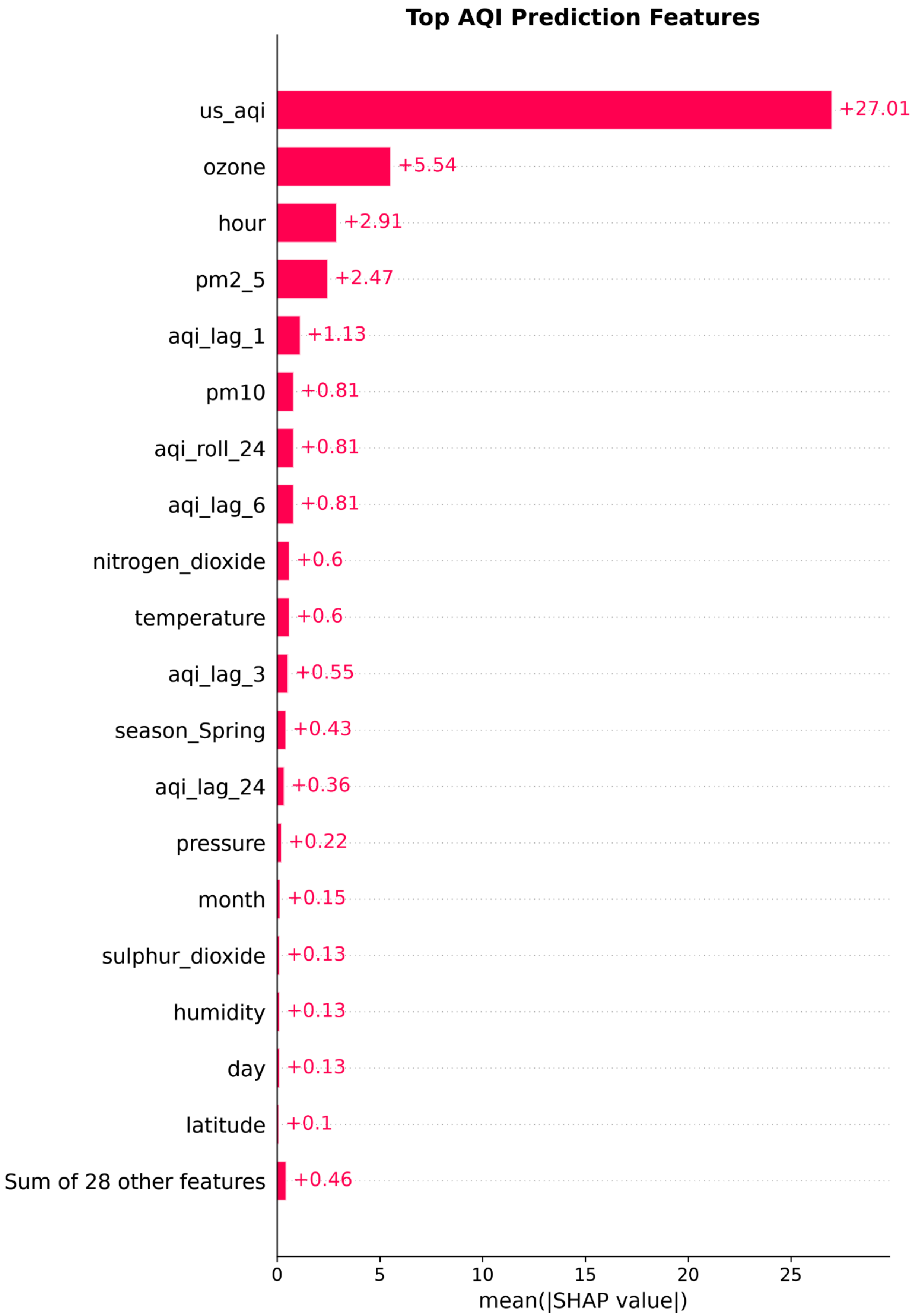

Top AQI Prediction Features
us_aqi
+27.01
ozone
+5.54
hour
+2.91
pm2_5
+2.47
aqi_lag_1
+1.13
pm10
+0.81
aqi_roll_24
+0.81
aqi_lag_6
+0.81
nitrogen_dioxide
+0.6
temperature
+0.6
aqi_lag_3
+0.55
season_Spring
+0.43
aqi_lag_24
+0.36
pressure
+0.22
month
+0.15
sulphur_dioxide
+0.13
humidity
+0.13
day
+0.13
latitude
+0.1
Sum of 28 other features
+0.46
0
5
10
15
20
25
mean(|SHAP value|)

Figure 16. Example local SHAP explanation for an individual AQI prediction.

11. Production Monitoring and Retraining

Monitoring evaluates feature-distribution change and predictive performance. In the validated run, 35.71% of monitored numeric features were identified as drifting, with a mean Population Stability Index (PSI) of 0.809088. Model performance remained strong with RMSE 6.76914, MAE 4.19394, and R^2 0.96636.

Because meaningful performance degradation was not observed, the retraining decision was negative despite detected feature drift. This prevents unnecessary replacement of a well-performing production model.

Metric	Observed Value
Numeric features with drift	35.71%
Mean PSI	0.809088
RMSE	6.76914
MAE	4.19394
R^2	0.96636
Retraining required	No

Table 8. Validated production monitoring result.

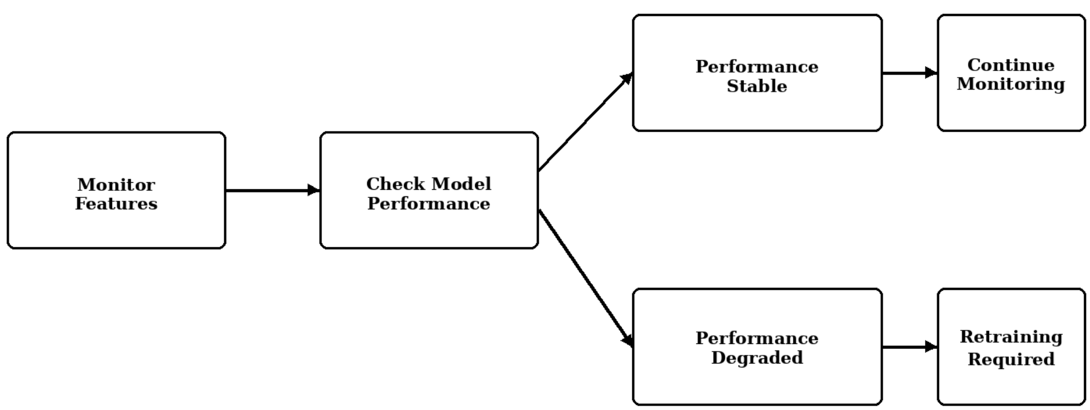


Figure 17. Monitoring and conditional retraining decision flow.

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MONITORING SUMMARY

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Overall Status : DRIFT_DETECTED

Production Model : LightGBM

Production Version : v1

Feature Drift

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Numeric Features : 28

Drift Features : 10

Moderate Drift : 10

No Drift Features : 8

Drift Percentage : 35.71%

Mean PSI : 0.809088

Categorical Drift

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Features Checked : 3

Drift Features : 1

Moderate Drift : 0

Model Performance

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Current MAE : 4.19393551981666

Current RMSE : 6.769140106297623

Current R^2 : 0.9663587320437662

Registered RMSE : 6.7691

RMSE Alert Threshold : 30.0

Performance Status : NORMAL

WARNING:

Significant feature drift was detected.

Retraining assessment should be performed.

12. FastAPI Backend and Testing

FastAPI provides programmatic access to the production system. The implemented routes support service health, model metadata, supported cities, current conditions, city-specific forecasts, forecast requests, and refresh operations.

Method	Endpoint	Purpose
GET	/	Service information
GET	/health	Health status
GET	/model	Production model metadata
GET	/cities	Supported cities
GET	/current/{city}	Current city conditions
GET	/forecast/{city}	City forecast
POST	/forecast	Forecast request
POST	/refresh	Refresh operation

Table 9. Main FastAPI endpoints.

The API test suite was updated to reflect the implemented behavior. The validated run completed with 12 passing tests. Remaining warnings relate to FastAPI event-handler deprecation and do not prevent operation.

Figure 18. FastAPI prediction service.

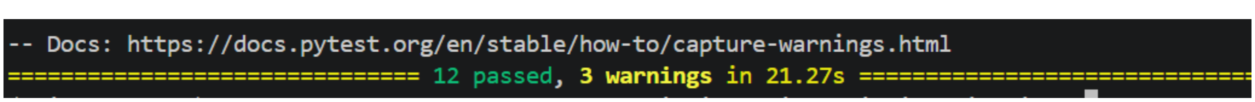


Figure 19. FastAPI automated test execution showing all 12 API tests passing successfully.

13. Streamlit Dashboard and User Experience

The Streamlit dashboard is the primary user-facing component of the system. The final version was redesigned using a professional dark-themed interface with a fixed sidebar, clearer information hierarchy, responsive Plotly visualizations, and explicit user interaction. The user selects one of the ten supported cities and presses the Generate Dashboard button. A progress indicator is displayed while current conditions, the 72-hour AQI forecast, and explainability resources are prepared.

The dashboard displays a Current AQI overview with an AQI gauge and category status, live weather information, current pollutant concentrations, and health advice. The forecast chart is labelled Next 24-Hour AQI Forecast to clearly distinguish predicted values from historical observations. Dynamic city-specific y-axis scaling is used so small but meaningful forecast variations remain visible.

A 3-Day AQI Forecast summarizes forecast conditions by day using predicted AQI values and AQI-category styling. Decorative sun and cloud icons were removed because the current system does not independently forecast future weather conditions. Weather and pollutant values are persisted during recursive forecasting, so the interface avoids presenting unsupported future-weather information.

The final dashboard also includes Model Explainability, Smart Activity Planner, and City Comparison functionality. Model Explainability displays local SHAP contributions for the next AQI prediction. Smart Activity Planner converts forecast AQI values into practical activity recommendations and identifies more suitable time windows. City Comparison evaluates forecast statistics across all supported cities, including near-term AQI, 24-hour average, 72-hour average, and peak forecast AQI.

Sidebar navigation separates the application into Dashboard, Forecast, City Comparison, Smart Planner, Model Explainability, Data & Insights, and About pages. Existing features such as hazardous-AQI alerts, health guidance, detailed forecast tables, and CSV export are retained.

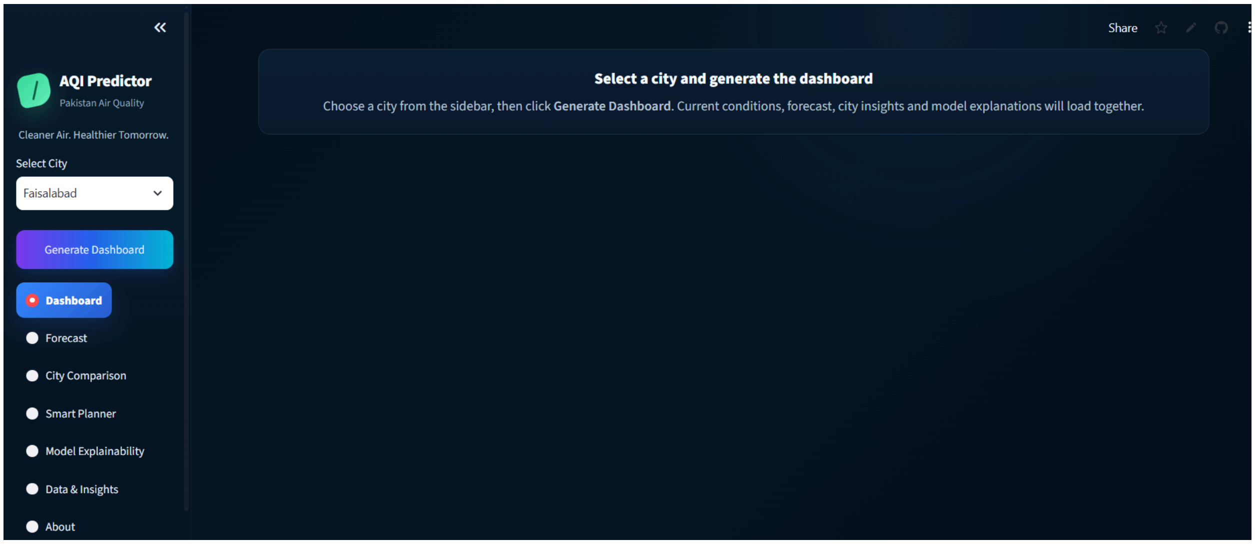


Figure 20. Deployed Streamlit AQI dashboard.

Figure 21. dashboard showing Current AQI, weather, health advice, and pollutant conditions.

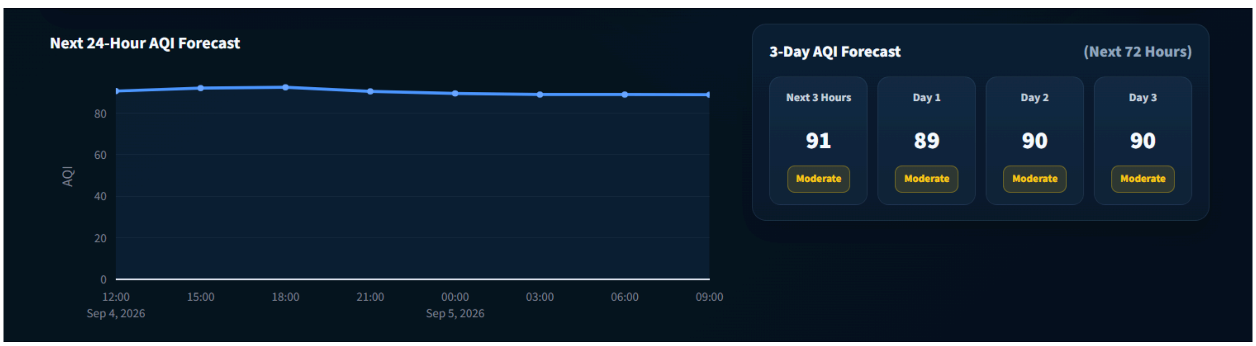


Figure 22. Next 24-Hour AQI Forecast and 3-Day AQI Forecast for the selected city.

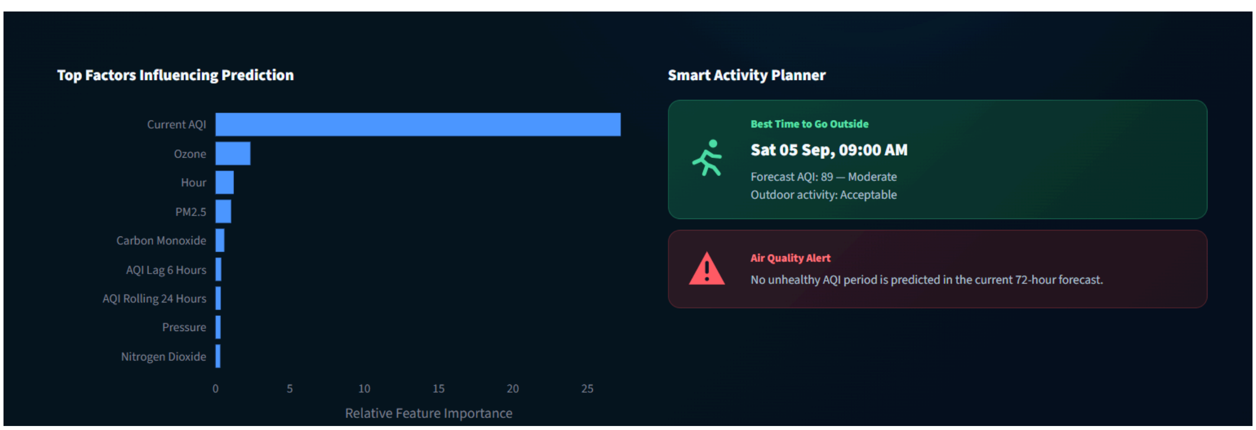


Figure 23. Dashboard-integrated SHAP feature importance and Smart Activity Planner providing forecast-based outdoor activity guidance.

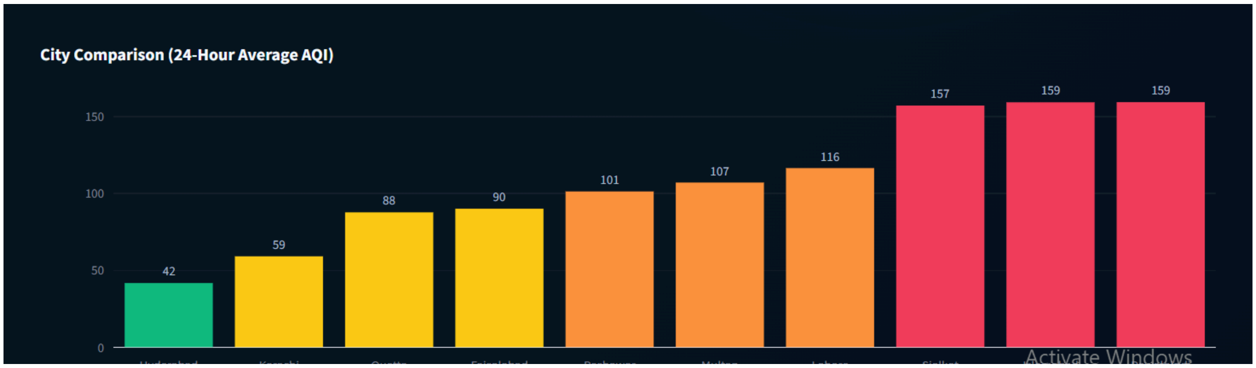


Figure 24. Main-dashboard comparison of the 24-hour average forecast AQI across supported Pakistani cities.

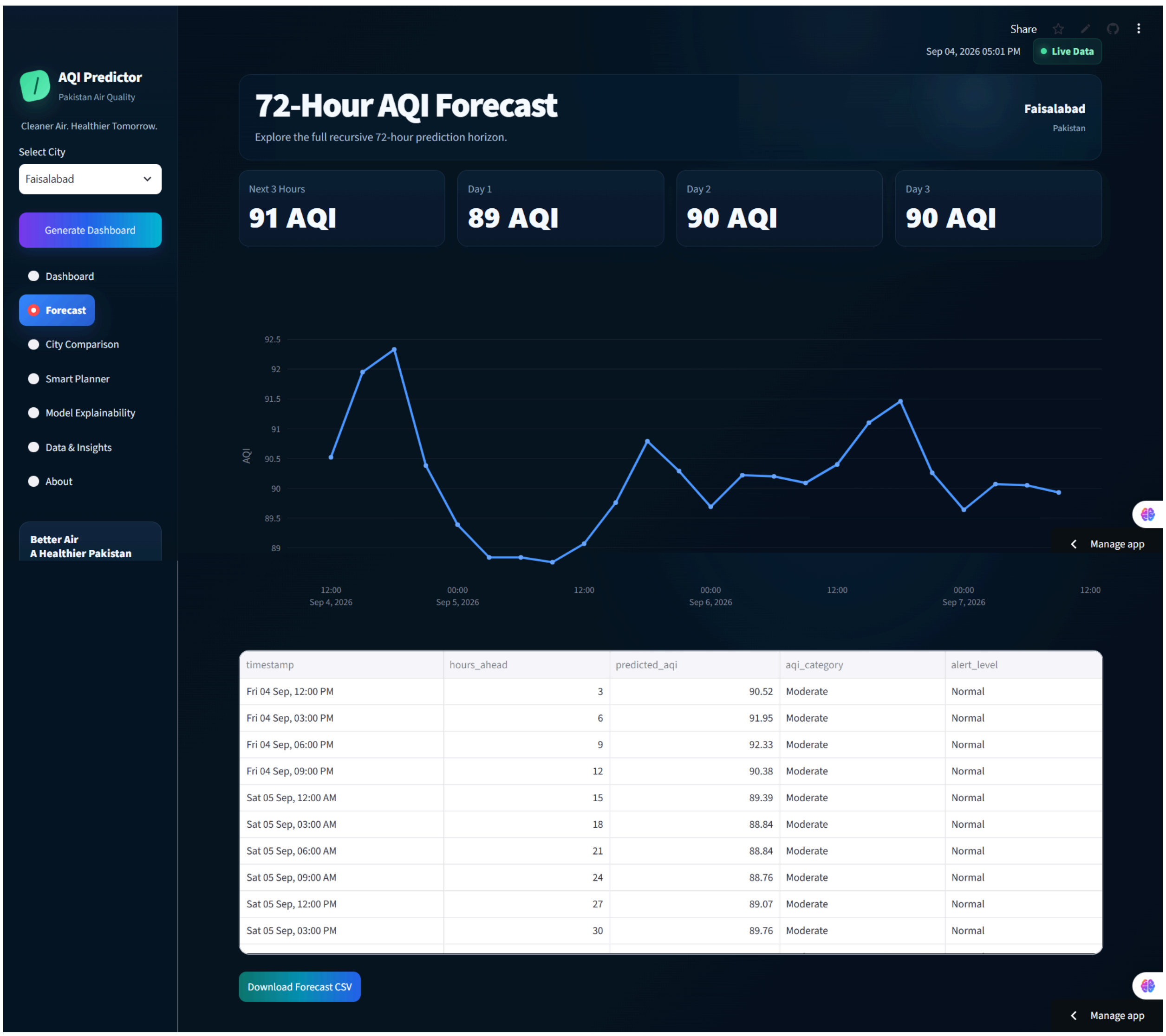
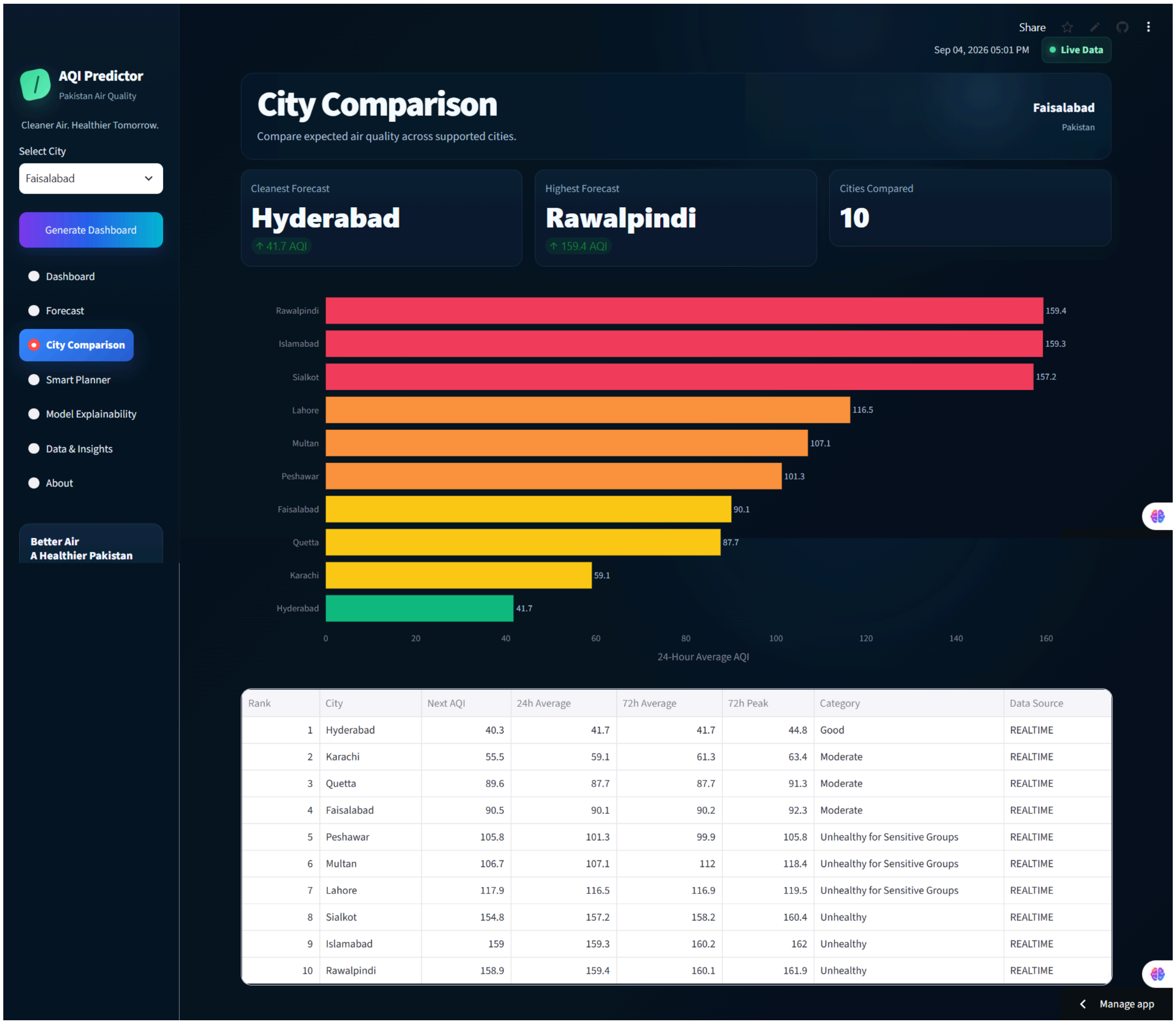


Figure 25. Dedicated 72-hour AQI forecast page showing forecast summaries, three-hour prediction intervals, AQI categories, alert levels, and downloadable forecast results.



Rank	City	Next AQI	24h Average	72h Average	72h Peak	Category	Data Source
1	Hyderabad	40.3	41.7	41.7	44.8	Good	REALTIME
2	Karachi	55.5	59.1	61.3	63.4	Moderate	REALTIME
3	Quetta	89.6	87.7	87.7	91.3	Moderate	REALTIME
4	Faisalabad	90.5	90.1	90.2	92.3	Moderate	REALTIME
5	Peshawar	105.8	101.3	99.9	105.8	Unhealthy for Sensitive Groups	REALTIME
6	Multan	106.7	107.1	112	118.4	Unhealthy for Sensitive Groups	REALTIME
7	Lahore	117.9	116.5	116.9	119.5	Unhealthy for Sensitive Groups	REALTIME
8	Sialkot	154.8	157.2	158.2	160.4	Unhealthy	REALTIME
9	Islamabad	159	159.3	160.2	162	Unhealthy	REALTIME
10	Rawalpindi	158.9	159.4	160.1	161.9	Unhealthy	REALTIME

Figure 26. Dedicated City Comparison page ranking the ten supported cities using 24-hour average forecast AQI and detailed 72-hour forecast statistics.

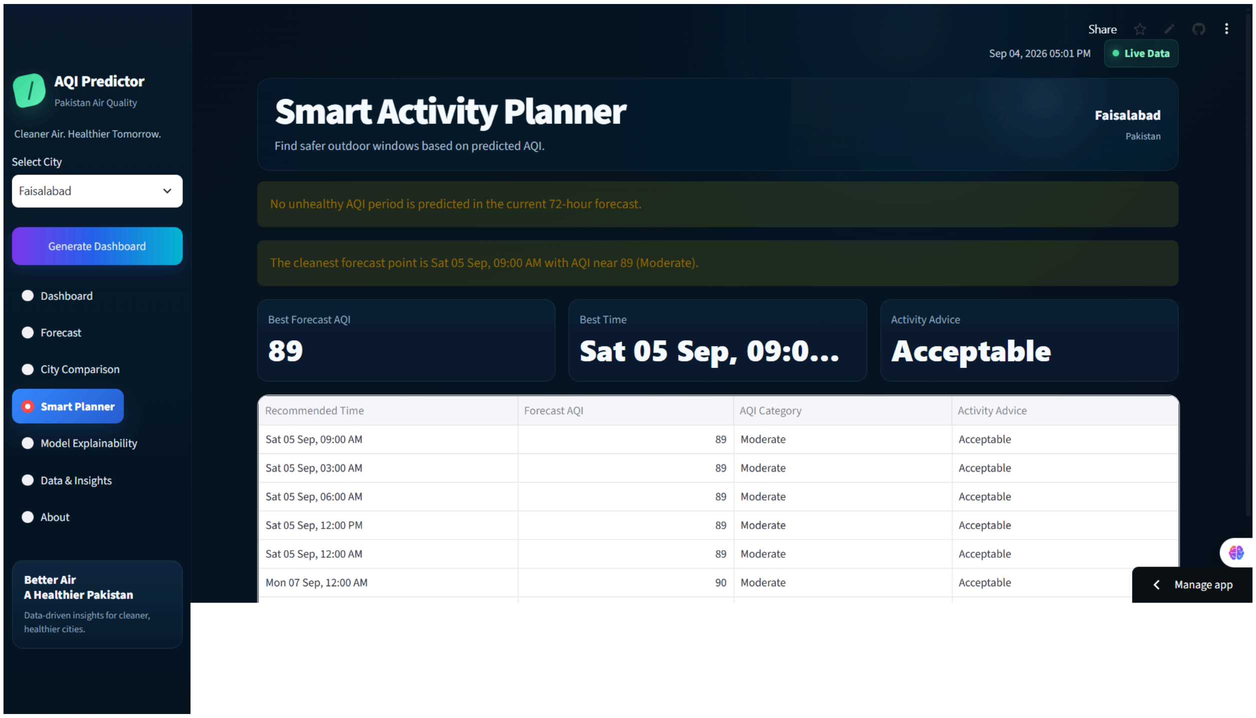


Figure 27. Smart Activity Planner identifying safer outdoor activity windows and providing AQI-based activity recommendations from the 72-hour forecast.

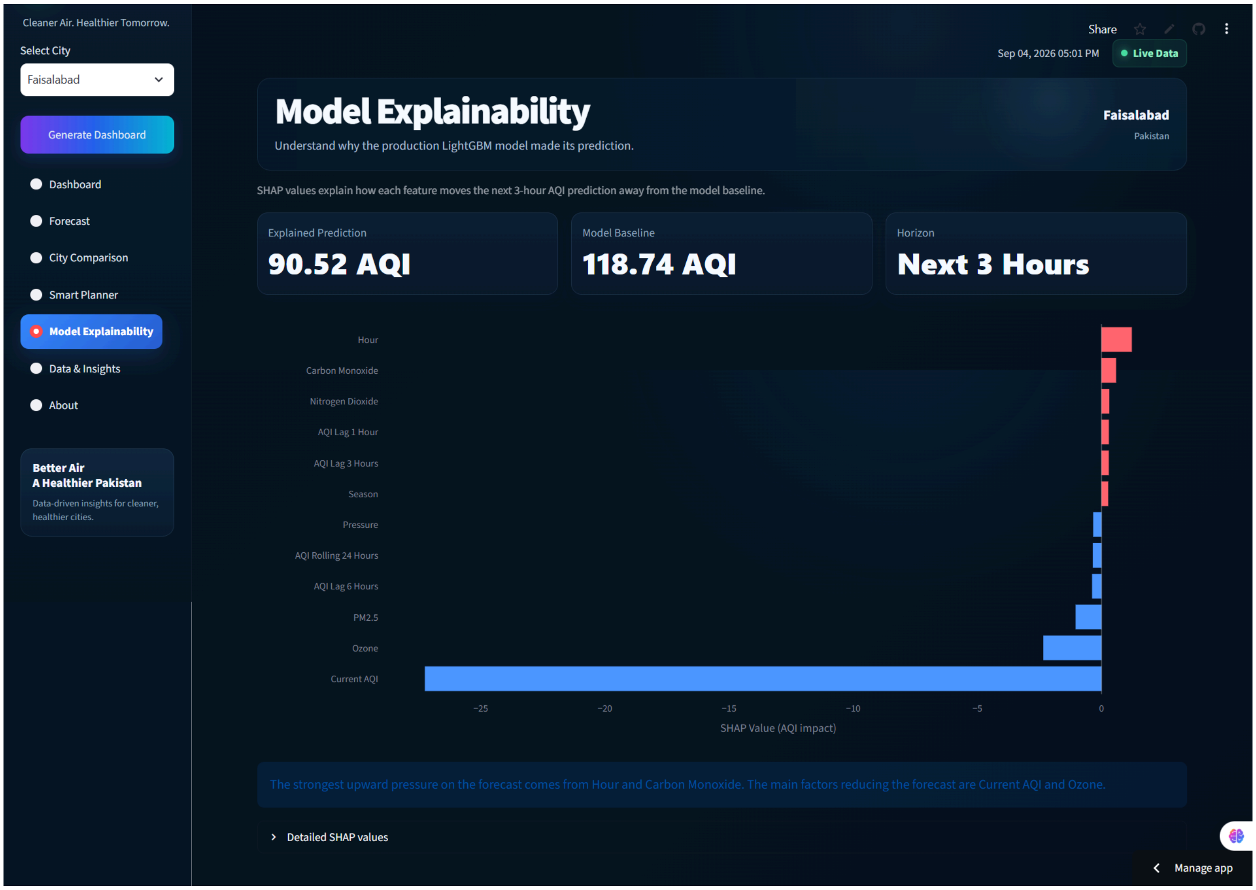


Figure 28. Model Explainability page showing local SHAP feature contributions to the next three-hour AQI prediction.

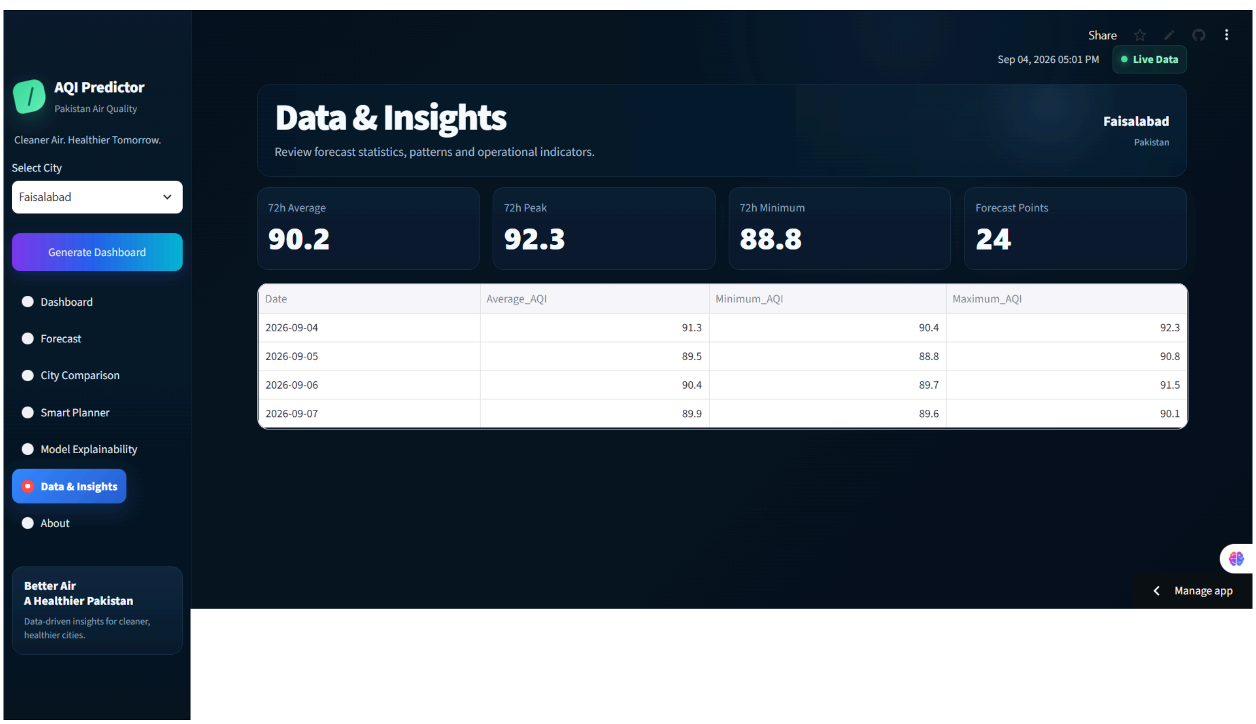


Figure 29. Data & Insights page summarizing the 72-hour forecast average, peak, minimum, forecast-point count, and daily AQI statistics.

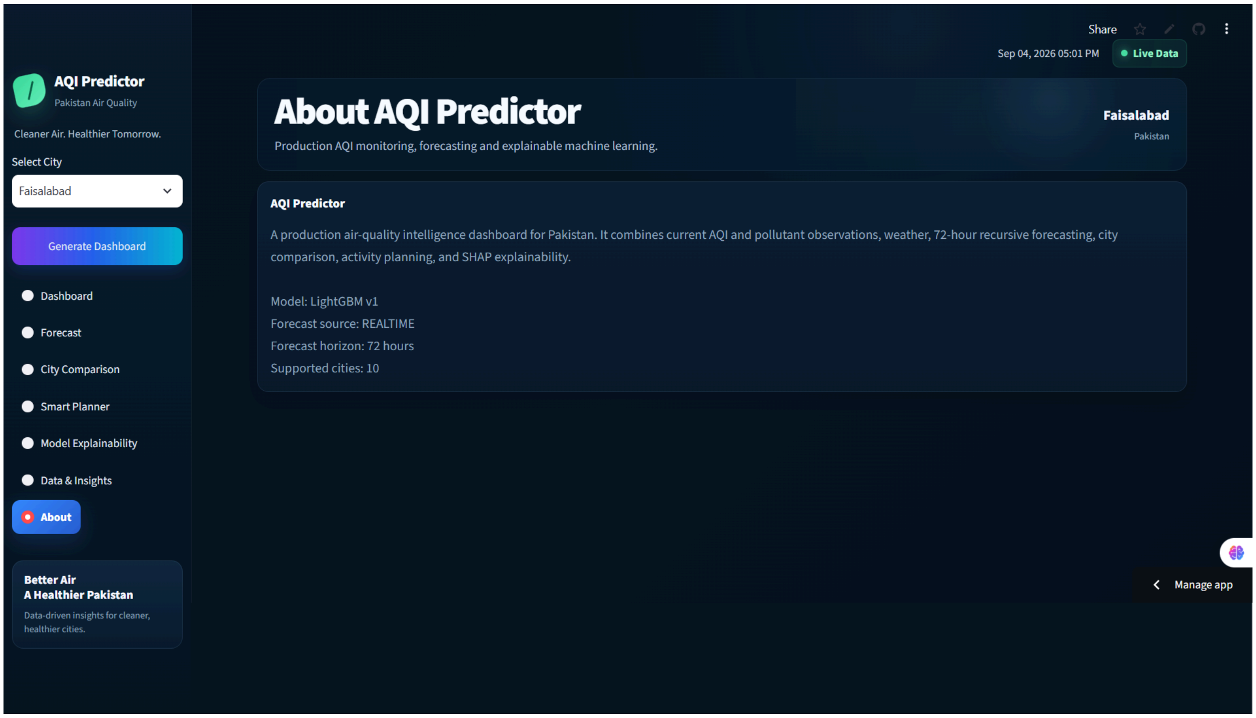


Figure 30. About page summarizing the AQI Predictor system, production LightGBM model, real-time forecast source, 72-hour prediction horizon, and supported cities.

14. AQI Classification and Hazardous Alerts

Forecasts are converted into AQI categories and operational alert levels so users do not need to interpret numerical values alone. AQI values above 300 are classified as Hazardous and assigned an Emergency alert. If any prediction in the 72-hour horizon exceeds this threshold, the dashboard displays an explicit hazardous-condition warning and corresponding health guidance.

AQI Range	Category	Operational Response
0-50	Good	Normal conditions
51-100	Moderate	Moderate conditions
101-150	Unhealthy for Sensitive Groups	Sensitive-group caution
151-200	Unhealthy	Warning
201-300	Very Unhealthy	High Alert
>300	Hazardous	Emergency

Table 10. AQI classification and operational alert mapping.

15. Automation and Cloud Deployment

15.1 GitHub Actions

The repository contains separate GitHub Actions workflows for hourly pipeline execution, daily monitoring, and retraining-related operations. Automation allows real-time data to accumulate and monitoring tasks to execute without manual intervention.

Workflow	Purpose
hourly_pipeline.yml	Collect/update real-time data and maintain feature history
daily_monitoring.yml	Evaluate production drift and model performance
retraining_pipeline.yml	Execute retraining-related workflow when required

Table 11. GitHub Actions workflows.

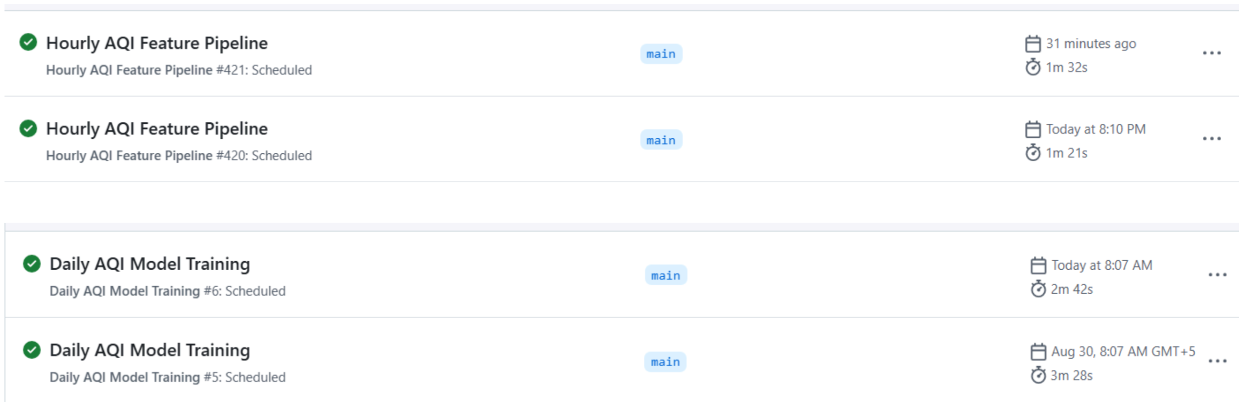


Figure 31. Automated GitHub Actions workflow execution.

15.2 Streamlit Community Cloud

The dashboard was deployed on Streamlit Community Cloud using Python 3.11. External service credentials are provided through deployment secrets rather than committed source code. The final application runs the production forecasting logic used during local development.

During cloud deployment, the dashboard initially encountered a module import error because Streamlit executed the application from the dashboard directory. This was resolved by adding the project root to the Python import path before importing modules from the src package. The deployment also includes SHAP as a runtime dependency so local explainability can be generated directly in the hosted application.

16. Challenges and Solutions

Challenge	Resolution
Insufficient early real-time history	Validated coverage and used historical fallback until enough real-time observations accumulated.
Forecast chart appeared visually flat	Introduced dynamic city-specific y-axis scaling and removed zero-anchored area filling.
Streamlit Cloud import error	Added the project root to sys.path before importing src modules.
Initial cloud runtime incompatibility	Redeployed using Python 3.11 to match the supported project environment.
Alternative provider required card verification	Used Streamlit Community Cloud for the user-facing deployment.
Automated commits caused Git divergence	Used stash/pull/rebase/push workflow when bot-generated commits existed.
Forecasting approaches used different protocols	Reported results with explicit evaluation limitations rather than claiming a direct ranking.
Feature drift without performance loss	Used conditional retraining based on drift and performance instead of drift alone.

Table 12. Main implementation challenges and resolutions.

17. Internship Achievements and Learning Outcomes

17.1 Main Achievements

- Prepared a validated 43,170-row historical dataset for ten Pakistani cities.
- Implemented historical and real-time feature pipelines with Hopsworks integration.
- Completed EDA covering city, temporal, pollutant, correlation, category, and hazardous-AQI analysis.
- Evaluated seven conventional machine-learning models and selected LightGBM for production.
- Implemented ARIMA and LSTM experiments to cover statistical and deep-learning forecasting.
- Built a recursive 72-hour forecast pipeline producing 24 predictions per city.
- Implemented SHAP global and local explanations.
- Created a versioned model registry and monitoring with drift and performance evaluation.
- Implemented conditional retraining logic, FastAPI services, and GitHub Actions automation.
- Validated the API with 12 passing tests.
- Developed a Streamlit dashboard with health guidance, hazardous alerts, and forecast export.
- Successfully deployed the application on Streamlit Community Cloud.
- Implemented Smart Activity Planner functionality using forecast AQI to provide practical activity recommendations.
- Implemented City Comparison to compare predicted air-quality conditions across all ten supported cities.

17.2 Learning Outcomes

The project provided practical experience across the complete machine-learning lifecycle. Key learning outcomes include constructing time-dependent features without leakage, comparing models with appropriate metrics, separating offline experiments from production requirements, managing model artifacts, interpreting model behavior with SHAP, and designing monitoring logic that considers both input drift and predictive performance.

The implementation also developed software-engineering and MLOps skills, including modular Python organization, API design, automated testing, dependency management, Git/GitHub workflows, CI/CD automation, cloud deployment, secret management, and user-interface refinement. These activities demonstrated that reliable machine-learning systems depend on the surrounding data and software infrastructure as much as on the predictive model.

18. Limitations and Future Improvements

18.1 Current Limitations

- The historical dataset covers approximately six months, limiting representation of full annual seasonality.
- Islamabad and Rawalpindi contain identical AQI values across comparable historical timestamps.
- Recursive forecasting can propagate earlier prediction errors into later steps.
- Future exogenous pollutant and weather values are persisted during recursive forecasting rather than independently forecast at every horizon step.
- ARIMA, LSTM, and LightGBM metrics originate from different evaluation protocols and are not a strict apples-to-apples benchmark.

- The current registry and monitoring design is appropriate for project scale but could be expanded for larger multi-model production environments.
- Smart Activity Planner recommendations are based on AQI thresholds and forecast values and should be interpreted as general guidance rather than personalized medical advice.

18.2 Future Improvements

- Extend the historical dataset to multiple years to capture broader seasonal and extreme-pollution patterns.
- Integrate future weather forecasts and, where available, pollutant estimates into multi-step forecasting.
- Evaluate direct multi-horizon or sequence-to-sequence forecasting to reduce recursive error accumulation.
- Add calibrated forecast uncertainty or prediction intervals.
- Expand automated monitoring dashboards and registry promotion/rollback controls.
- Add more cities and independent data sources to improve geographical coverage and validation.

19. Requirement Compliance

Requested Requirement	Status	Evidence
EDA to identify trends	Completed	City, monthly, hourly, correlation, category, and hazardous-AQI analyses.
Variety of forecasting models	Completed	Seven ML models, ARIMA baseline, and LSTM experiment.
SHAP or LIME explainability	Completed	Global and local SHAP outputs for LightGBM.
Hazardous AQI alerts	Completed	Hazardous category, Emergency alert, health guidance, and dashboard warning.
Automated pipeline operations	Completed	Hourly, daily monitoring, and retraining GitHub Actions workflows.
Interactive dashboard	Completed	Streamlit dashboard with current conditions, 72-hour forecasting, next-24-hour visualization, local SHAP explainability, Smart Activity Planner, City Comparison, health guidance, hazardous alerts, and forecast export.
API/backend	Completed	FastAPI endpoints and 12 passing tests.
Detailed final report	Completed	Implementation, results, challenges, limitations, and evidence placeholders documented here.

Table 13. Project requirement compliance.

20. Conclusion

This internship project delivered an end-to-end AQI monitoring and forecasting system for ten Pakistani cities. The solution combines historical and real-time environmental data, engineered temporal features, multiple forecasting approaches, a versioned LightGBM production model, recursive 72-hour prediction, SHAP explainability, monitoring, conditional retraining, FastAPI services, GitHub Actions automation, hazardous-AQI alerts, and a deployed Streamlit dashboard.The final deployed dashboard extends the system beyond basic forecast visualization by providing local SHAP explanations, Smart Activity Planner recommendations, multi-city comparison, and a more professional interactive user experience.

LightGBM was selected from seven conventional models with RMSE 6.7691, MAE 4.1939, and R^2 0.9664. ARIMA and LSTM provided statistical and deep-learning experiments without replacing the production model because their evaluation and operational characteristics differ. The final system therefore demonstrates not only predictive modelling but a maintainable MLOps workflow from data processing to user-facing deployment.